

(~4–8.5 standard drinks) alcohol ingestion after strenuous resistance or high-intensity interval training impedes exercise-induced phosphorylation of the mammalian target of rapamycin complex 1 (mTORC1) pathway and reduces rates of muscle protein synthesis, an effect that also appears to be most prominent in men (60, 165). Alcohol also has known diuretic effects, causing an increase in excess urine production of about 10 milliliters for every gram of ethanol ingested (63). Therefore, hydration-related effects are most significant when concentrated alcohol solutions are consumed, and consumption of dilute alcohol solutions or alcohol with adequate additional fluid intake can offset the diuretic effect (191). In addition, large alcohol doses can dramatically affect thermoregulatory ability, augmenting environmental heat loss at low ambient temperatures (88, 89).

Finally, alcohol can impair exercise performance and recovery via effects on sleep and autonomic function. For example, evening binge alcohol consumption (4–5 standard drinks within 2 hours) disrupts sleeping cardiac vagal tone and reduces sleep quality (53, 93), and increases sympathoexcitation and disrupts blood pressure regulation the following morning (94). These disruptions may serve to increase cardiovascular strain and impair performance during subsequent exercise, although this has not been directly tested. It is important to note that many of these effects are likely dependent on the amount of alcohol consumed, and studies to tease out dose-dependent effects (particularly at high doses) are difficult both logistically and ethically.

Evidence also shows that chronic alcohol consumption can impair muscle metabolism and impede training-induced adaptations. Skeletal muscle from middle-aged men and women with chronic alcohol abuse displays increased catabolic markers (e.g., MuRF-1, ubiquitinated proteins), decreased anabolic markers (e.g., phosphorylated 4E-BP1), and decreased insulin receptor substrate 1, which plays a critical role in transmitting signals from insulin and IGF-1 to coordinate cellular growth and metabolism (189, 190). Therefore, chronic alcohol consumption can impede training-related adaptations and promote signatures consistent with myopathy in muscle, but the degree to which these effects occur are likely directly related to the degree of alcohol use (or abuse).

Adverse Effects

β -Blockers may also have an **ergolytic** effect (reduce performance). Studies have shown that β -blockers impair the cardiovascular response to exercise by reducing maximal heart rate, oxygen consumption, and 10-kilometer (6.2 miles) race time performance (6). Also, β -blockers are associated with an increased rating of perceived exertion (205). Risks associated with these drugs include bronchospasm, heart failure, prolonged hypoglycemia, bradycardia, heart block, and intermittent claudication (78).

Dietary Supplements

The global market for sports nutrition products continues to grow at a significant pace, with an estimated market value of over US\$53.6 billion (208). Some sports nutrition companies make unsubstantiated claims concerning the efficacy of their products. At times, unscrupulous companies have knowingly placed analogs of banned substances in their products to enhance their effects (44). As a result, much confusion has been created among athletes regarding appropriate supplements and the ethics of companies. This section describes dietary supplements that are commonly used by athletes and presents scientific examination of their efficacy.

Essential Amino Acids, Branched-Chain Amino Acids

Essential amino acids (EAAs) are not produced in the body and must be obtained through the diet. They include histidine, isoleucine, leucine, valine, lysine, methionine, phenylalanine, threonine, and tryptophan. High levels of EAAs can be found in any number of animal-based proteins or as supplements sold over the counter. Leucine can also be found in plants such as spinach and broccoli, but the amount of leucine is minimal when compared to that in animal-based sources.

Efficacy

Scientists have been able to delineate certain categories of amino acids as well as individual amino acids and their role in stimulating muscle protein synthesis. For example, Tipton and colleagues (206) reported that nonessential amino acids were not necessary to stimulate muscle protein synthesis; rather, only the presence of EAAs is needed. In this research (206), six healthy adults, three males and three females, participated in resistance exercise (8 sets of 8 repetitions at 80% 1RM) and then consumed either 40 grams of mixed amino acids (composed of both nonessential and essential

amino acids), 40 grams of EAA, or a placebo. While the mixed amino acids and the EAA induced an anabolic response that was significantly greater than with the placebo treatment, no difference was observed between the amino acid treatments. Since there was no difference, it was concluded that nonessential amino acids are not needed to stimulate muscle protein synthesis. In a follow-up study, using the same protocol (but with only 6 g EAA and 35 g sugar), Rasmussen and colleagues (180) demonstrated significantly greater anabolic drive—that is, the building of new muscle tissue—with the EAA supplement (when given shortly after resistance exercise) versus the placebo.

Subsequently, Tipton and colleagues (207) examined the effects of consuming 6 grams EAA plus 36 grams of sugar, before or after resistance training, on muscle protein metabolism. They reported that when the EAA plus sugar was consumed 30 minutes before resistance training, the acute (3 hr after exercise) anabolic response was 158% greater than when the EAA and sugar supplement were consumed after training. As a result of these acute findings, Tipton and colleagues (207) and Rasmussen and colleagues (180) theorized that a person who consumed EAA before or after (or both before and after) every resistance training session over a period of weeks would experience greater changes in muscle mass than with training only. It is important to note that although the nonessential amino acids are not required to stimulate muscle protein synthesis, this does not mean that they are not important for maximizing training adaptations in the athlete. The conditionally essential and nonessential amino acids serve as the substrates that are incorporated into the newly formed muscle proteins; further, they also spare the conversion of these amino acids from the essential ones, thus effectively increasing the levels of EAAs.

Prior work in animal models further identified that of the eight EAAs, it is the **branched-chain amino acids (BCAAs)**—isoleucine, leucine, and valine—that are responsible for increasing muscle protein synthesis (83). Of the BCAAs, it appears that leucine is the

key amino acid for stimulating muscle protein synthesis (126) via the Akt/mechanistic target of rapamycin (mTOR) pathway ([figure 12.4](#)). Norton and coworkers (159) demonstrated in rodent models that a leucine threshold for stimulating muscle protein synthesis exists. Based on this pioneering research, many nutrition researchers now believe that the leucine content in any given protein intake is the rate-limiting factor in terms of maximizing muscle protein synthesis within the capacity of skeletal muscle rather than simply the total amount of protein ingested. To support this theory of the importance of the content of a protein meal, Pasiakos and colleagues (166) reported that a leucine-enriched EAA beverage resulted in a significant elevation of muscle protein synthesis compared to a normal leucine-containing EAA beverage. In this study, trained males ingested one of two different 10-gram EAA beverages during cycling exercise—one containing 3.5 grams of leucine (46 mg/kg body mass) and the other containing 1.87 grams of leucine (~25 mg/kg body mass). After the exercise bout (during a 3-hr assessment period), muscle protein synthesis was 33% greater in the leucine-enriched EAA beverage. The importance of leucine intake and its role in the anabolic response of skeletal muscle following resistance training continues to be a popular topic in sports nutrition research today.

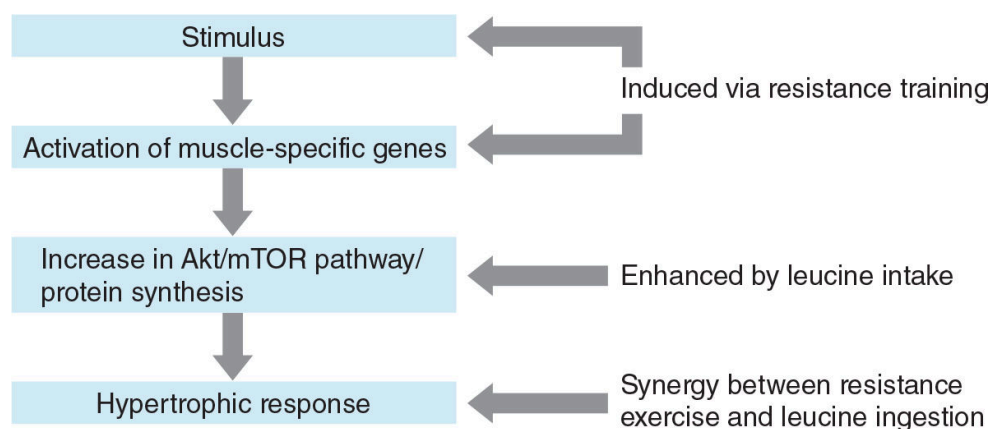


FIGURE 12.4 Role of leucine in muscle protein synthesis.

The branched-chain amino acid leucine is a key regulator in stimulating muscle protein synthesis. Leucine directly activates the Akt/mTOR pathway in skeletal muscle, which is a key pathway in skeletal muscle protein synthesis.

Adverse Effects

No known reports or scientific studies have investigated the adverse effects of branched-chain amino acid or EAA supplementation.

Arginine

Arginine is a conditionally essential amino acid and has crucial roles in nutrition and metabolism. Arginine is required for the synthesis of protein and creatine, and its metabolism results in the production of nitric oxide. The claims often attributed to arginine supplementation are its ability to elevate nitric oxide levels, increase muscle blood flow, and improve exercise performance. Each of these claims is directed at athletes or physically active individuals, because these are all outcomes that would be beneficial for enhancing sport performance or maximizing training adaptations. Unfortunately, there is very little scientific evidence supporting these claims regarding arginine supplementation in the populations that these outcomes would benefit.

Efficacy

Oral arginine supplementation is most often marketed for its potential to elevate nitric oxide levels. Nitric oxide possesses many physiological roles in the human body, but its effects on **vasodilation** (i.e., the widening of blood vessels) are what make it important during exercising conditions. During exercise, nitric oxide levels are naturally increased so that more blood can flow through the arteries and arterioles for the purpose of delivering oxygen and fuel substrates to the working skeletal muscles (27). While nitric oxide is essential for vasodilation, the overwhelming majority of the

scientific literature does not support the claim that oral arginine supplementation increases nitric oxide production to levels that are greater than the effects of exercise alone in healthy individuals (5, 142).

A natural outcome of nitric oxide production is increased muscle blood flow. Therefore, increasing muscle blood flow is also a popular claim that is made regarding arginine's effectiveness. Despite the common perception among athletes and consumers of dietary supplements that arginine supplementation does increase muscle blood flow, investigations into this area are in agreement that muscle blood flow is not increased in healthy individuals following arginine supplementation (203). Oral arginine supplementation has also been purported to improve endurance exercise performance. However, in research investigating arginine's effectiveness in healthy populations, it was reported that time to exhaustion, local muscular endurance, and intermittent anaerobic exercise performance were not improved as a result of supplementation (95, 142, 214). Due to its inability to increase nitric oxide production, enhance muscle blood flow, or improve several modes of endurance exercise, oral arginine supplementation is not recommended for healthy athletes.

Adverse Effects

A majority of the scientific investigations in which arginine was given to participants to improve exercise performance used a 6-gram dose. While this dose of arginine was not effective in improving endurance performance, this amount was well tolerated and posed no adverse side effects. Oral supplementation at doses up to 13 grams is generally well tolerated. The most common adverse reactions to higher doses (13-30 g) include gastrointestinal distress consistent with nausea, abdominal cramps, and diarrhea (45).

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β -Hydroxy- β -Methylbutyrate

β -Hydroxy- β -methylbutyrate (HMB) is a derivative of the EAA leucine and its metabolite α -ketoisocaproic acid. Evidence shows that HMB plays a role in the regulation of muscle protein breakdown in the body (decreases protein breakdown) by inhibiting the ubiquitin–proteasome pathway (193). Due to HMB’s role in the regulation of protein breakdown, it may be an effective supplement for minimizing losses of lean muscle mass in situations that promote a catabolic state. β -Hydroxy- β -methylbutyrate is not currently banned or restricted by sport organizations.

Efficacy

The initial research study that highlighted HMB’s anticatabolic potential was conducted by Nissen and colleagues (158). In this study, untrained subjects ingested one of three levels of HMB (0, 1.5, or 3.0 g per day) and two protein levels (117 or 175 g per day) and resistance trained three days per week for three weeks. As a measure of myofibrillar protein breakdown, urinary 3-methyl-histidine levels were measured (higher levels of urinary 3-methyl-histidine mean that more muscle protein is catabolized, or broken down). After the first week of the resistance training protocol, urinary 3-methyl-histidine was increased by 94% in the control group and by 85% and 50% in those individuals ingesting 1.5 and 3 grams of HMB per day, respectively. During the second week of the study, urinary 3-methyl-histidine levels were still elevated by 27% in the control group but were 4% and 15% below basal levels for the 1.5 and 3 grams of HMB per day groups. At the end of the third week of resistance training, urinary 3-methyl-histidine levels were not significantly different between the groups (158).

Research in other catabolic-inducing environments has also demonstrated the effectiveness of HMB supplementation. For

example, when older adults were confined to complete bed rest (a catabolic state) for 10 days, it was reported that the subjects who were supplemented with 3 grams of HMB per day lost significantly less lean body mass (lost only 0.37 lb [0.17 kg]) as compared to those subjects who were supplemented with a placebo (lost 4.5 lb [2.04 kg]) (55). Other research findings are in agreement with these observations and have shown that HMB exerts an anticatabolic effect and suppresses muscle damage (130).

In previously untrained individuals initiating a resistance training program lasting four to eight weeks, HMB has been shown to exert significantly greater improvements in strength and lean body mass as compared to a placebo (158, 164). The muscle damage and soreness that result from resistance training are predictably greater in those who have little training experience as compared to individuals who have engaged in resistance training for longer periods of time. Given HMB's ability to suppress muscle damage and muscle protein breakdown, it is not surprising that HMB is effective for people beginning a new training program.

The ergogenic effects of HMB in trained individuals are less conclusive. Studies using resistance-trained or competitive athletes were unable to duplicate the results seen in a recreationally trained population using similar supplementation schedules (160, 179). However, many of the studies investigating the effects of HMB on strength and lean body mass in trained individuals were of short duration (less than five weeks) and lacked the presence of a periodized, high-intensity training program. In order for HMB to be effective, it may be essential to provide novel stimuli to individuals to induce either muscle damage or stimulate elevated protein breakdown. It appears that trained individuals instructed not to change their programs do not benefit from HMB. Investigations that have provided an adequate training stimulus in trained individuals have also reported increases in maximal strength and lean body mass as compared to placebo treatments (158, 164).

Adverse Effects

The duration, dosage, and timing of HMB supplementation have notably varied in the scientific literature. In nearly every published investigation related to HMB supplementation, 3 to 6 grams per day was ingested. Three grams per day (often divided into several doses) is the most common dosage used in published studies. The majority of studies investigating HMB have used the calcium salt of HMB—HMB-Ca. Another form of HMB, HMB-free acid, has been investigated as well (192). The safety of HMB supplementation has been widely studied, and currently, the consensus is that HMB presents no known adverse effects.

β -Hydroxy- β -methylbutyrate is most effective when an adequate training stimulus is provided. For untrained individuals, this does not likely require high-volume training. For trained individuals, a high-intensity, high-volume resistance training program is likely needed for benefits to be realized with HMB supplementation.

Nutritional Muscle Buffers

During high-intensity anaerobic exercise, a significant accumulation of hydrogen ions (H^+) is coupled with a reduction of pH within skeletal muscle and has been shown to adversely affect performance (182). The ability to regulate H^+ concentration in skeletal muscle during high-intensity exercise has been termed **muscle buffering capacity (MBC)**. There is a strong positive relationship between MBC and exercise performance (repeated-sprint ability, high-intensity exercise capacity, anaerobic threshold, and training volume) (62). In fact, researchers have demonstrated a positive relationship between exercise performance and MBC in athletes who participate in sports such as basketball, soccer, hockey, cycling, crew (rowing), triathlons, and sprinting (62). In theory, improving MBC by training or nutritional means (β -alanine, sodium bicarbonate, or citrate) would

improve performance in sports and activities that may be limited by H⁺ buildup. Therefore, a brief review of β -alanine, sodium bicarbonate, and sodium citrate and their effects on high-intensity exercise performance follows.

β -Alanine

β -Alanine is a nonessential amino acid that is common in many foods that we eat, such as chicken. By itself, β -alanine has limited ergogenic properties. However, in muscle cells, it is the rate-limiting substrate for carnosine synthesis (59). Harris and colleagues (104) reported that four weeks of supplementing with β -alanine (4-6 g/day) resulted in a mean increase of 64% in skeletal muscle β -alanine concentrations. In humans, carnosine is found primarily in fast-twitch (type II) skeletal muscle and is estimated to contribute up to 40% of the skeletal MBC of H⁺ produced during intense anaerobic exercise, thus encouraging a drop in pH (104, 109). Theoretically, increasing skeletal muscle carnosine levels through chronic training or β -alanine supplementation (or both) would improve MBC and most likely improve anaerobic performance. Interestingly, carnosine concentrations in athletes such as sprinters and bodybuilders appear to be significantly higher than those in marathoners, untrained individuals, and older people (104, 202).

Suzuki and colleagues (202) examined the relationship between skeletal muscle carnosine levels and high-intensity exercise performance in trained cyclists. The authors reported a significant and positive relationship between carnosine concentration and mean power in a 30-second maximal sprint on a cycle ergometer. This finding supported the theory that skeletal muscle carnosine levels have a positive correlation with anaerobic performance because of the relationship between carnosine and MBC.

Efficacy

β -Alanine has been studied for its effects on strength, aerobic power, and high-intensity short-term exercises interspersed with short recovery intervals. In contrast to creatine, β -alanine does not seem

to improve maximal strength (127). Similarly, aerobic power does not appear to be improved with β -alanine supplementation (127). Even though aerobic power is not improved, supporting data indicate that anaerobic threshold is improved with β -alanine supplementation (226). Practically, improving anaerobic threshold (as measured by the lactate and ventilatory thresholds) means that endurance activities can be performed at relatively higher intensities for longer periods. Hill and colleagues (109) examined the effect of β -alanine supplementation on muscle carnosine levels and exercise performance in untrained subjects. In a double-blind fashion, 25 male subjects (19-31 yr) supplemented with either 4.0 grams β -alanine or sugar placebo for the first week, then with up to 6.4 grams for an additional nine weeks. Muscle carnosine levels (via muscle biopsy) and total work done (kilojoules) were measured at weeks 0, 4, and 10 during cycling to exhaustion at maximal power established from a graded exercise cycle ergometry test. Mean carnosine levels increased by 58% at week 4 and an additional 15% at week 10. Additionally, 13% and 16% increases in total work done during cycle ergometry were seen at weeks 4 and 10, respectively.

In a comprehensive review summarizing the effects of β -alanine supplementation on high-intensity performance, Artioli and colleagues (9) stated that β -alanine ingestion is capable of improving performance in exercises resulting in an extreme intramuscular acidotic environment, such as multiple bouts of high-intensity exercises lasting more than 60 seconds, as well as single bouts undertaken when fatigue is already present. High-intensity exercises performed with a lower level of acidosis are unlikely to benefit from β -alanine supplementation.

Adverse Effects

In the published literature, β -alanine ingestion has ranged from 2.4 to 6.4 grams per day. In many β -alanine trials, the total daily amount of β -alanine ingestion was divided into two to four smaller doses. The reason for the smaller dosing strategies is to prevent the only reported adverse effect of β -alanine supplementation, which is

the symptom of paresthesia (tingling, pricking, or numbness of a person's skin) (9). Symptoms of paresthesia are triggered by a high and acute single dose and disappear within approximately 1 hour after the ingestion (9, 104).

Sodium Bicarbonate

Sodium bicarbonate is an antacid (alkalinizing agent), meaning that it counteracts or neutralizes acid (low pH). Sodium bicarbonate is naturally formed in the body and is also found in baking soda. Supplementation with sodium bicarbonate has been shown to increase the pH of blood (198). A pH difference is created between the inside and outside of muscle cells that causes an accelerated movement of H^+ out of the contracting muscle, helping to regulate intramuscular pH. Supplementing with sodium bicarbonate has been shown to improve MBC and in turn high-intensity exercise performance.

Efficacy

Most research investigating the efficacy of sodium bicarbonate for enhancing sport performance has focused on short bouts of high-intensity exercise lasting 60 seconds to 6 minutes (150). McNaughton and colleagues (148) have demonstrated improved short-term, high-intensity work from acute sodium bicarbonate supplementation. Hobson and colleagues (111) examined the effects of sodium bicarbonate ingestion during a 2,000-meter (2,187 yd) rowing ergometer time trial in experienced male rowers. The rowers were supplemented before exercise with 0.3 grams per kilogram body mass of sodium bicarbonate or a placebo. Time to complete the 2,000 meters and time taken for each 500-meter (547 yd) split were recorded. While there were no significant differences in the overall 2,000-meter time trial, the sodium bicarbonate did result in significant improvements in the third and final 500-meter time splits. Some investigations (26, 140), but not all (10), have reported similar improvements in high-intensity exercise following 0.3 grams per kilogram body mass sodium bicarbonate supplementation. The

timing of the sodium bicarbonate ingestion used in many scientific investigations is typically 60 to 90 minutes before the bout of exercise.

Adverse Effects

It appears that a dosage of 0.3 grams per kilogram body mass (136 mg/lb body mass) sodium bicarbonate ingested approximately 60 to 180 minutes before activity improves short-duration, high-intensity exercise performance (97, 111, 150). Dosages above this amount have been associated with unpleasant side effects such as diarrhea, cramping, nausea, and vomiting. A more tolerable dose (0.2 g/kg [91 mg/lb] body weight) has been shown to reduce these side effects; however, research has revealed that this dose does not improve exercise performance (139). It appears that a sodium bicarbonate dose ranging from 0.3 to 0.5 grams per kilogram body mass, 60 to 180 minutes before exercise, is optimal for improvement of performance (97). Due to the severity of the side effects (which may be experienced by some athletes even when ingesting 0.3 g/kg body mass), many sport scientists recommend that athletes try sodium bicarbonate supplementation during practice before using it as a precompetition aid.

Sodium Citrate

Although sodium citrate is not actually a base, it can increase blood pH without the gastrointestinal distress that is commonly seen with sodium bicarbonate supplementation (211). It is believed that once in the blood, sodium citrate actually breaks down into bicarbonate, thus increasing the extracellular pH. As a result, sodium citrate would help regulate intramuscular pH during high-intensity exercise by the same mechanism as sodium bicarbonate.

Efficacy

The data are equivocal in terms of the effectiveness of sodium citrate's ergogenic potential during short-duration, high-intensity exercise, with some investigations reporting no benefit (13, 211) and

others having observed an ergogenic effect (141, 149). For example, a dose of 200 milligrams per pound body mass (0.44 g/kg body weight) given 60 to 90 minutes before exercise led to significant improvement (about 20% greater) in leg muscular endurance during maximal isometric knee extensions (105). However, more research is needed before sodium citrate can be recommended before a workout or competition for performance enhancement.

Adverse Effects

Sodium citrate may possess MBC and has been used without the typical gastrointestinal discomfort usually associated with sodium bicarbonate ingestion (13, 141). Even though it appears that the adverse effects of sodium citrate supplementation are lower than those of sodium bicarbonate, ingestion of 0.4 to 0.6 grams per kilogram body mass sodium citrate has the potential to cause gastrointestinal distress (161). The gastrointestinal response seems to vary between individuals; therefore, athletes should individually test sodium citrate supplementation before using it in competition.

L-Carnitine

L-carnitine is synthesized from the amino acids lysine and methionine and is responsible for the transport of fatty acids from the cytosol into the mitochondria to be oxidized for energy. Carnitine's role in lipid oxidation has generated interest in its efficacy as a dietary supplement, primarily to enhance exercise performance by increasing fat use and sparing muscle glycogen. However, studies examining L-carnitine's role as an ergogenic aid for increasing lipid oxidation and subsequently enhancing performance are inconsistent (85). Although Bacurau and colleagues (11) showed enhanced fatty acid oxidation following three weeks of L-carnitine supplementation, which was attributed to a greater carnitine content in the muscle, most studies have been unable to demonstrate elevated muscle carnitine levels following supplementation (218). This may be related to limits in the amount of carnitine that can be absorbed through oral supplementation, or potentially related to limits in the amount of

fat that can be transported into the mitochondria through the carnitine system due to feedback regulators within the muscle, such as malonyl–coenzyme A, which is a product of metabolism.

Efficacy

Interestingly, several studies have suggested that L-carnitine may enhance recovery from exercise (119, 216). Decreases in pain and muscle damage (84), decreases in markers of metabolic stress (195), and enhanced recovery (216) have also been demonstrated following high-intensity resistance exercise in untrained or recreationally trained individuals who supplemented with L-carnitine. The mechanisms that have been proposed involve enhancing blood flow regulation through an enhanced vasodilatory effect that reduces the magnitude of exercise-induced hypoxia (120). In addition, Kraemer and colleagues indicated that L-carnitine supplementation (2 g/day for three weeks) upregulates androgen receptors (132) and increases IGF binding proteins that preserve IGF-1 concentrations (133). These endocrine adaptations from the supplement may have an important role in the enhanced recovery seen following high-intensity exercise.

Adverse Effects

Up to 3 grams of daily L-carnitine supplementation (for three weeks) appears to be well tolerated in healthy volunteers, with no adverse subjective, hematological, or metabolic events reported (183). However, as with most supplements, this information should not be extrapolated to suggest the safety of greater dosing or of use for prolonged supplementation periods.

Creatine

Creatine is a nitrogenous organic compound that is synthesized naturally in the body, primarily in the liver, and helps to supply energy to all cells in the body. It can also be synthesized in smaller amounts in both the kidneys and pancreas. The amino acids

arginine, glycine, and methionine are the precursors for the synthesis of creatine in those organs. Creatine can also be obtained through dietary sources. It is found in relative abundance in both meat and fish. Approximately 98% of creatine is stored within skeletal muscle in either its free form (40%) or its phosphorylated form (60%). Smaller amounts of creatine are also stored in the heart, brain, and testes. Creatine is transported from its site of synthesis to the skeletal muscle via circulation.

Importance of Creatine to Exercise

Creatine, in the form of creatine phosphate (CP; also called phosphocreatine [PCr]), has an essential role in energy metabolism as a substrate for the formation of adenosine triphosphate (ATP) by rephosphorylating adenosine diphosphate (ADP), especially during short-duration, high-intensity exercise. The ability to rapidly rephosphorylate ADP is dependent on the enzyme creatine kinase and the availability of CP within the muscle. As CP stores become depleted, the ability to perform high-intensity exercise declines. In short-duration sprints (e.g., a 100 m sprint), the energy for fueling the activity is derived primarily through the hydrolysis of CP. However, as high-intensity exercise increases in duration, the ability of CP to serve as an energy source is drastically reduced.

Depletion of muscle CP during high-intensity exercise is the primary mechanism leading to fatigue in such events. During a 6-second bout of maximal exercise, CP levels within the muscle are reduced 35% to 57% from resting levels (81). As the duration of high-intensity exercise increases to 30 seconds, CP levels in the muscle are further reduced by approximately 64% to 80% from resting levels (28, 40); during bouts of repeated high-intensity exercise, CP levels in the muscle are almost completely depleted (147). As muscle CP concentrations decrease, the ability to perform maximal exercise is reduced. Hirvonen and colleagues (110) demonstrated that sprint times were slower as CP concentrations were reduced. It stands to reason that if muscle CP concentrations

could be maintained, the ability to sustain high-intensity exercise would be improved. This is the basis for creatine supplementation in athletes.

Creatine Supplementation

Survey-based studies suggest that the prevalence of creatine use among athletes and military personnel is about 15% to 40%, with use more common in male strength or power athletes (134). As a result of their widespread use, creatine supplements may be the most extensively studied ergogenic aid in recent history. Creatine supplementation is reported to increase the creatine content of muscles by approximately 20% (65, 121). However, it appears that a saturation limit for creatine exists within muscle. Once creatine concentrations in skeletal muscle reach 150 to 160 mmol per kilogram dry weight, additional supplementation appears to be unable to further increase muscle creatine concentrations (92). This has important implications for the “more is better” philosophy that governs the thinking of some athletes and may affect the development of proper and realistic dosing strategies.

A typical creatine supplementation regimen involves a loading dose of 20 to 25 grams daily for five days, or 0.3 grams per kilogram body mass if an individual wishes to dose relative to body weight, followed by a maintenance dose of 2 grams per day (121). If one ingests creatine without an initial loading dose, muscle creatine content will reach levels similar to those seen in people who do initially use a loading dose, but reaching that muscle creatine concentration will take longer (~30 days vs. 5 days). Muscle creatine levels will remain elevated as long as the maintenance dose is maintained (2 g/day or 0.03 g/kg body mass per day) (121). Once creatine supplementation is stopped, muscle creatine levels will return to baseline levels in approximately four weeks (65, 121).

While most of the available research involving creatine supplementation has focused on skeletal muscle, a small body of accumulating research has focused on creatine and potential

cognitive benefits. Creatine supplementation can increase brain creatine content in humans, but higher and more prolonged dosing strategies than those typically used to increase muscle creatine levels are required to increase brain creatine (57). At this time, the optimal dosing strategies needed to induce this response are currently unknown, but this is an active area of research. It appears that creatine is most likely to exert an influence in situations where cognitive processes are stressed (e.g., during sleep deprivation, experimental hypoxia, and cognitively demanding tasks). Furthermore, creatine has shown some promise for attenuating symptoms of concussion, mild traumatic brain injury, and depression (72).

Efficacy

Most studies examining the effect of creatine supplementation on strength performance have been fairly consistent in showing significant ergogenic benefits ([figure 12.5](#)) (217). Strength increases in the bench press, squat, and power clean may be two to three times higher in trained athletes supplementing with creatine compared to placebo (113). These results may highlight the benefit of creatine supplementation in experienced resistance-trained athletes whose potential to improve strength may be limited. In experienced strength athletes, supplementing with creatine may also enhance the quality of workouts (less fatigue, enhanced recovery), which may be crucial for providing a greater training stimulus to the muscle.

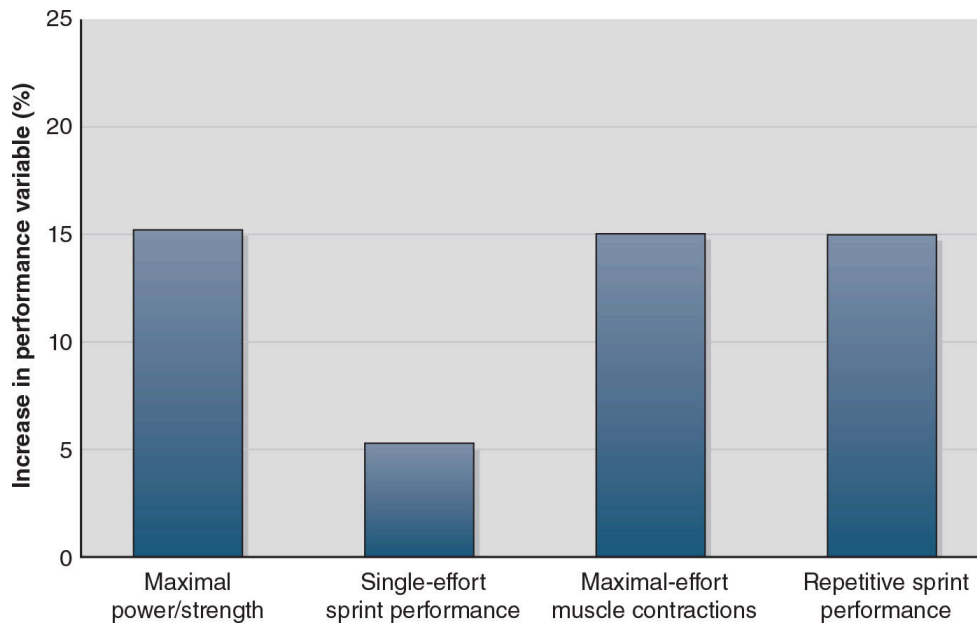


FIGURE 12.5 Approximate changes in performance variables following creatine supplementation.

Data from Wax et al. (2021).

Creatine supplementation has been shown to increase maximal strength, power, and lean body mass in both trained and untrained populations. Additionally, creatine supplementation is safe and relatively inexpensive.

Most studies examining the effects of creatine supplementation on a single bout of explosive exercise (sprint performance) have not shown any significant performance improvements (129, 155). However, in many of these studies, subjects were supplemented with a loading dose for only five days (129, 155). While ingesting creatine with a loading dose for five days will significantly increase intramuscular creatine stores, this is likely not enough time for training adaptations to occur and to be manifested in explosive exercise performance improvements. When subjects have supplemented for an extended period of time (28-84 days), significant improvements in jump and power performance have been

seen (215). It appears that creatine is more effective as a training supplement than as a direct performance enhancer.

Body Mass Changes

Prolonged creatine supplementation has been generally associated with increases in body weight. These increases appear to primarily relate to increases in fat-free mass. The increase in body mass is believed to be partly related to an increase in total body water. The water retention phenomenon occurs in the early stages (first several days) of supplementation (102). An increase in creatine content within the muscle is thought to enhance the intracellular osmotic gradient, causing water to fill the cell (8). In general, creatine supplementation during training is typically associated with a 1- to 4-pound (0.5-1.8 kg) greater increase in body mass or fat-free mass (71) over a period of several weeks to several months.

Adverse Effects

Increase in body mass could be a potential unwanted side effect of creatine supplementation, particularly in relation to strength-to-mass or power-to-mass ratio. However, in athletes supplementing with creatine, weight gain is often a desired outcome. When people talk about side effects from a drug or supplement, they are generally referring to a potentially debilitating effect. There have been many anecdotal reports of gastrointestinal, cardiovascular, and muscular problems, including muscle cramps, in association with creatine ingestion. However, controlled studies have been unable to document any significant side effects from creatine supplementation. Even during prolonged supplementation (several weeks up to many months), no increases in reported side effects were noted in subjects supplementing with creatine—either competitive athletes or recreationally trained individuals (8).

Another concern with any supplement is the long-term health effects. A retrospective study on health variables in 26 former or current competitive athletes who had used creatine for up to four

years indicated only occasional gastrointestinal upset during the loading phase (185). These disturbances ranged from gas to mild diarrhea. Other hypothesized major concerns during creatine supplementation include a strain on the kidneys due to the high nitrogen content of creatine and increased creatine excretion during short-term ingestion. However, no renal dysfunction has been reported with either short-term (five days) or long-term (up to five years) creatine use (170, 171).

Despite certain media and anecdotal reports linking creatine supplementation to dehydration and cramping, there is no reason to believe that creatine enhances the risk for these side effects.

Stimulants

There is little question about the usefulness of stimulants to many types of athletic endeavors; the effects of stimulants include reduced fatigue, increased alertness, increased confidence, and even euphoria. Because stimulants have to be used at the time of competition to confer an ergogenic advantage, drug testing performed immediately after the competition can effectively reduce their abuse (118). However, many stimulants (e.g., caffeine) are also available in food items, while others (e.g., ephedrine) may be found in decongestants and other medications.

Caffeine

Caffeine is a multifaceted supplement used in athletics and is popular among both aerobic endurance and resistance training athletes. It is found in coffee, tea, soft drinks, chocolate, and various other foods. It is a central nervous system stimulant, and its effects are similar to, yet weaker than, those associated with amphetamines. Caffeine has been used as a performance enhancer for more than 40 years and is one of the few ergogenic aids that may be used comparably by aerobic and anaerobic athletes.

The mechanism to explain endurance improvements is unclear. One of the mechanisms that has been proposed involves an increase in fat oxidation through the mobilization of free fatty acids from adipose tissue or intramuscular fat stores (90). The greater use of fat as a primary energy source slows glycogen depletion and delays fatigue. Alterations in muscle metabolism alone cannot fully explain the ergogenic effect of caffeine. During short-duration, high-intensity exercise, the primary ergogenic effect attributed to caffeine supplementation is enhanced power production. This is thought to be the result of an enhanced excitation–contraction coupling, affecting neuromuscular transmission and mobilization of intracellular calcium ions from the sarcoplasmic reticulum (90).

Efficacy

For the aerobic athlete, caffeine is thought to prolong aerobic endurance exercise (221). Initial studies dealing with the effect of caffeine supplementation on aerobic endurance performance indicated a 21-minute improvement in the time to exhaustion (from 75 min in a placebo trial to 96 min in the caffeine trial) during cycling at 80% of $\dot{V}O_2\text{max}$ (49). These results were confirmed by other investigations demonstrating the ergogenic effect of caffeine during prolonged aerobic endurance activity (17). These studies showed that caffeine in doses ranging from 3 to 9 milligrams per kilogram (equivalent to approximately 1.5 to 3.5 cups of automatic drip coffee in a 70 kg [154 lb] person) produced a significant ergogenic effect. Similar performance effects from caffeine have also been demonstrated during short-duration, high-intensity exercise (34, 58).

When the effects of caffeine ingestion on sprint or power performance are examined, however, the ergogenic benefits become less clear. Several studies have shown that caffeine ingestion does not improve power performance (46, 96), but those studies used recreational athletes. In elite judo athletes ingesting caffeine (5 mg/kg), it was reported that peak power was significantly improved in comparison with a placebo treatment (194).

Similarly, in competitive swimmers, caffeine ingestion (250 mg) was shown to improve sprint times during repeated 100-meter (109 yd) sprints by an average of 3% (47). The number of studies on caffeine ingestion and sprint and power performance is limited compared to the number that measured caffeine's effect on aerobic endurance performance. Thus, the results of the studies on the ergogenic benefit of caffeine for power performance are inconclusive. If any performance benefit occurs, it is likely in the trained athlete.

Caffeine from a food source (coffee consumption) and anhydrous caffeine have both shown significant performance effects; however, the extent of performance improvements appears to be greater when caffeine is ingested in tablet form (91). When caffeine is provided as a pure caffeine supplement, the ergogenic benefit with respect to improved aerobic endurance performance has been reported to range between 28% and 43% (91). However, when it is provided in a food source such as coffee (either caffeinated coffee or decaffeinated coffee with added caffeine), the ergogenic benefit for aerobic endurance exercise may not be seen (91). Graham and colleagues (91) suggested that although the bioavailability of caffeine is the same whether it is consumed in a food source or in an anhydrous form, some compound in coffee antagonizes the action of caffeine.

In doses of 3 to 9 milligrams per kilogram body weight consumed approximately 60 minutes before exercise or during prolonged exercise, caffeine is ergogenic. No further benefit is observed when it is consumed at higher dosages (≥ 9 mg/kg). It can decrease feelings of perceived exertion, improve work capacity, and increase mental alertness. The scientific literature does not support caffeine-induced diuresis during exercise or any harmful change in fluid balance that would negatively affect performance.

Adverse Effects

The side effects associated with caffeine include anxiety, gastrointestinal disturbances, restlessness, insomnia, tremors, and heart arrhythmias. Caffeine is also physically addicting, and discontinuation can result in some withdrawal symptoms, such as headache, fatigue, dysphoric mood, difficulty concentrating, and flu-like somatic symptoms (200). Caffeine intakes greater than 9 milligrams per kilogram appear to result in a greater risk for side effects (99). Caffeine overdose is an issue that can have potentially fatal consequences. A lethal dose of caffeine is typically in excess of 5 grams, which equates to about 42 cups of coffee at 120 milligrams of caffeine per cup (128) or twenty-five 200-milligram tablets of caffeine.

Preworkout Energy Drinks

Energy drinks were initially introduced to the market in the United States in the 1990s and have since become one of the biggest selling functional beverages in the industry with annual sales projected to surpass \$21 billion by the year 2026 (123). The primary reasons for ingesting energy drinks include enhancing workouts, improving sport performance, and facilitating faster training adaptations (168). The ingredients most common to commercially available energy drinks are caffeine, taurine, ginseng, guarana, carbohydrates, B vitamins, tyrosine, and ginkgo biloba, among others. Of these ingredients, caffeine and carbohydrate are the primary ergogenic nutrients in energy drinks (123).

Efficacy

Preworkout energy drinks have been investigated for their effectiveness on several modes of exercise, including resistance exercise, anaerobic exercise, and aerobic endurance exercise. In relation to resistance exercise performance, Forbes and colleagues (70) gave 15 physically active college-aged students a commercially available energy drink or an isocaloric noncaffeinated placebo 60 minutes before exercise. The energy drink was standardized with 2

milligrams of caffeine per kilogram body mass. The exercise bout consisted of 3 sets of 70% 1RM bench press conducted to failure on each set with 1 minute of rest between each set. The energy drink significantly increased bench press performance as compared to the placebo treatment. Other studies investigating the effects of preworkout energy drink consumption have also reported improvements in upper and lower body total lifting volume (54, 86). While it appears that preworkout energy drink ingestion is effective for increasing resistance training volume performance, other types of anaerobic exercise (including Wingate tests and speed or agility performance) are not as responsive to energy drink consumption (70, 114). In terms of aerobic endurance exercise performance, some investigations (122), but not all (36), have reported that energy drinks containing approximately 2 milligrams of caffeine per kilogram body mass, consumed within an hour before exercise, improve cycling and running performance in trained cyclists and recreationally active individuals (123). Interestingly, the amount of caffeine (2 mg/kg body mass) in the energy drinks is less than the amount recommended for an enhancement of exercise performance (3-9 mg/kg body mass). This contributes to the hypothesis that the synergistic effects of the various ingredients in energy drinks may be responsible for the reported improvements in exercise performance (123).

Adverse Effects

The safety issues surrounding energy drink consumption appear related to excessive caffeine intake (187) and the detrimental effects of mixing energy drinks with alcohol (187). The mixing of energy drinks with alcohol is more prevalent in certain social contexts and is not a common practice for those consuming energy drinks before a workout to enhance work capacity or to facilitate faster training adaptations. Due to the presence of caffeine in most energy drinks, the same potential adverse effects that exist for caffeine also exist for energy drinks. However, most energy drinks contain only moderate amounts of caffeine, typically less than 300 milligrams

(187). Even though energy drinks typically contain low to moderate doses of caffeine, one should recognize that it is possible to overdose on caffeine through the consumption of energy drinks (116). While there is some information related to the side effects of excess caffeine ingestion, energy drinks contain other ingredients and stimulants. Currently, it is unknown whether inclusion of other stimulants in energy drinks may increase or decrease the threshold for experiencing side effects. Therefore, more research is needed to determine the long-term effects of habitual intake of energy drinks before definitive conclusions can be drawn.

Ephedrine

Another stimulant, ephedrine, has been widely used throughout the world for temporary relief from the symptoms of bronchial asthma, bronchitis, allergies, shortness of breath, cold and flu symptoms, and other ailments. Ephedrine is also very popular among bodybuilders due to its strong thermogenic quality. A thermogenic supplement or drug works by elevating basal metabolic rate, thereby increasing energy expenditure and ultimately resulting in fat loss. It often is used as a stacking agent with caffeine to enhance the thermogenic effect.

Efficacy

Studies examining the ergogenic effect of ephedrine have shown it to be effective only when it is taken in combination with caffeine (21). These results have been quite consistent with regard to improving aerobic endurance performance (20). In terms of anaerobic exercise performance, the scientific literature investigating the combination of ephedrine and caffeine is equivocal (224). However, side effects (vomiting and nausea) following exercise have been reported in 25% of the subjects ingesting a mixture of 5 milligrams per kilogram caffeine and 1 milligram per kilogram ephedrine (19). A subsequent study performed by the same investigators showed that at a lower dose (4 mg/kg caffeine and 0.8

mg/kg ephedrine), similar ergogenic benefits were realized with no side effects (20).

Adverse Effects

Ephedra, also called *ma huang*, is a plant that contains ephedrine. In the United States before 2004, most over-the-counter sport and weight-loss products contained ephedra. However, in April 2004, the FDA banned all products containing ephedra after determining that it posed an unreasonable risk to those who used it. The FDA ban was based on a report published by the Rand Institute (188) indicating that 16,000 adverse events were linked to the use of ephedra-containing dietary supplements. Further, Shekelle and colleagues (188) reported that the use of ephedra-containing dietary supplements or ephedrine plus caffeine was associated with an increased incidence of nausea, vomiting, psychiatric symptoms such as anxiety and change in mood, autonomic hyperactivity, palpitations, and, in a few cases, death. As a result of the many adverse effects, ephedrine use has been banned by most sport governing bodies, including the International Olympic Committee.

Citrus Aurantium

Citrus aurantium is from a fruit commonly known as *bitter orange* and is often used as an Asian herbal medicine to treat digestive problems (80, 100). However, it is also a mild stimulant and is thought to contribute to appetite suppression and increased metabolic rate and lipolysis (80, 100).

Efficacy

When citrus aurantium is combined with caffeine and other herbal products, significant improvements in time to fatigue have been reported (199). Citrus aurantium contains synephrine, a sympathomimetic agent, which some have suggested stimulates specific beta adrenergic receptors (β -3, but not β -1 or β -2) that in turn stimulate fat metabolism without any of the negative side effects generally associated with compounds that stimulate the other

adrenergic receptors (38). Synephrine, a known active component of citrus aurantium, is thought to interact with β -3 receptors to increase lipolysis (i.e., the breakdown of body fat) and minimize the cardiovascular effect typical of adrenergic amines (38). Currently, not enough data on citrus aurantium alone have been published to recommend its use as a performance-enhancing supplement.

Adverse Effects

Synephrine has been shown to stimulate peripheral α -1 receptors, resulting in vasoconstriction and elevations in blood pressure (31). However, other research has shown that when citrus aurantium is ingested alone, no effect on blood pressure is seen (103), but in combination with other herbal products it may cause significant elevations in systolic blood pressure (103). It should be noted that synephrine is currently on the National Collegiate Athletic Association's banned list of performance-enhancing drugs.

Summary of Ergogenic and Non-Ergogenic Dietary Supplements and Their Application

- **Essential amino acids and branched-chain amino acids.** Ability to stimulate muscle protein synthesis, particularly via the anabolic effects of the amino acid leucine
- **Arginine.** Not effective for inducing nitric oxide or muscle blood flow in healthy individuals; also does not appear to have any effect on improving endurance performance in healthy populations
- **β -Hydroxy- β -methylbutyrate (HMB).** Most effective for trained individuals participating in a high-intensity, high-volume resistance training program
- **β -Alanine.** Effective as a performance-enhancing substance in exercises inducing a high level of lactic acid, such as multiple bouts of high-intensity exercises lasting more than 60 seconds
- **Sodium bicarbonate.** Has been shown to improve high-intensity exercise performance and training but is associated with unpleasant side effects consistent with gastrointestinal discomfort, which would likely decrease performance

- ***Sodium citrate.*** Inconclusive scientific data in relation to sodium citrate's ergogenic potential during short-duration, high-intensity exercise
- ***L-carnitine.*** May enhance recovery following high-intensity resistance exercise
- ***Creatine.*** Effective for increasing maximal strength, power, lean body mass, and sprint performance in both trained and untrained populations; no other sport supplement is supported by the same level of positive research
- ***Caffeine.*** Effective for prolonging aerobic endurance exercise, but its effects on sprint or power performance are less clear
- ***Preworkout energy drinks.*** Effective for increasing resistance training volume performance and endurance performance but not ergogenic in their ability to improve anaerobic exercise (high-intensity cycling and speed or agility performance)
- ***Ephedrine.*** When combined with caffeine, is able to improve aerobic endurance performance; not as effective in improving anaerobic exercise performance
- ***Citrus aurantium.*** When combined with caffeine and other herbal products, improvements in exercise time to fatigue (a measure of endurance performance) have been reported

Conclusion

Athletes are frequently exposed to information about ergogenic aids. Strength and conditioning professionals play an integral part in the decision-making process regarding which ergogenic aids athletes choose to take. Therefore, it is imperative that strength and conditioning professionals stay up to date on the latest scientific findings on what ergogenic aids are safe, effective, and legal. If strength and conditioning professionals fail in this area, the athletes they serve will be more susceptible to hyped marketing, potentially false advertising, and influence by others who may not have the athlete's best interests in mind. The information contained in this chapter serves the strength and conditioning professional by providing fundamental information on performance-enhancing substances, including which substances are banned, effective or noneffective, and safe.

KEY TERMS

- acromegaly
- anabolic steroid
- androgenic
- β -agonists
- β -blockers
- β -hydroxy- β -methylbutyrate (HMB)
- branched-chain amino acids (BCAA)
- caffeine
- creatine
- dietary supplement
- ergogenic aid
- ergolytic
- erythropoietin (EPO)
- essential amino acids (EAA)
- human chorionic gonadotropin (HCG)

human growth hormone (HGH)
insulin
lipolysis
muscle buffering capacity (MBC)
muscle dysmorphia
peptides
random drug tests
selective androgen receptor modulators (SARMs)
stacking
testosterone
thermogenesis
vasodilation

STUDY QUESTIONS

1. Which of the following hormones is responsible for enhancing the ability to deliver oxygen to contracting skeletal muscle?
 - a. erythropoietin
 - b. insulin
 - c. human growth hormone
 - d. testosterone
2. What are the effects of evening binge alcohol consumption on sleep and autonomic function?
 - a. It improves sleep quality, increases cardiac vagal tone, and impairs blood pressure regulation.
 - b. It has no significant impact on sleep or autonomic function.
 - c. It disrupts sleeping cardiac vagal tone, reduces sleep quality, increases sympathoexcitation, and disrupts blood pressure regulation.
 - d. It improves blood pressure regulation and reduces sympathoexcitation during sleep.
3. What organization prohibits the use of SARMs in and outside of competition for all athletes?